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on writing

essays, volume three

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(in the process of answering an email, i accidentally wrote a tiny essay about writing. i usually spend weeks on an essay. this one took 67 minutes—23 of writing, and 44 of rewriting.)

i think it's far more important to write well than most people realize. writing doesn't just communicate ideas; it generates them. if you're bad at writing and don't like to do it, you'll miss out on most of the ideas writing would have generated.

as for how to write well, here's the short version: write a bad version 1 as fast as you can; rewrite it over and over; cut ~~out~~ everything unnecessary; write in a conversational tone; develop a nose for bad writing, so you can see and fix it in yours; imitate writers you like; if you can't get started, tell someone what you plan to write about, then write down what you said; expect 80% of the ideas in an essay to happen after you start writing it, and 50% of those you start with to be wrong; be confident enough to cut; have friends you trust read your stuff and tell you which bits are confusing or drag; don't (always) make detailed outlines; mull ideas over for a few days before writing; carry a small notebook or scrap paper with you; start writing when you think of the first sentence; if a deadline forces you to start before that, just say the most important sentence first; write about stuff you like; don't try to sound impressive; don't hesitate to change the topic on the fly; use footnotes to contain digressions; use anaphora to knit sentences together; read your essays out loud to see (a) where you stumble over awkward phrases and (b) which bits are boring (the paragraphs you dread reading); try to tell the reader something new and useful; work in fairly big quanta of time; when you restart, begin by rereading what you have so far; when you finish, leave yourself something easy to start with; accumulate notes for topics you plan to cover at the bottom of the file; don't feel obliged to cover any of them; write for a reader who won't read the essay as carefully as you do, just as pop songs are designed to sound ok on crappy car radios; if you say anything mistaken, fix it immediately; ask friends which sentence you'll regret most; go back and tone down harsh remarks; publish stuff online, because an audience makes you write more, and thus generate more ideas; print out drafts instead of just looking at them on the screen; use simple, germanic words; learn to distinguish surprises from digressions; learn to recognize the approach of an ending, and when one appears, grab it.

i try to write using ordinary words and simple sentences.

that kind of writing is easier to read, and the easier something is to read, the more deeply readers will engage with it. the less energy they expend on your prose, the more they'll have left for your ideas.

and the further they'll read. most readers' energy tends to flag part way through an article or essay. if the friction of reading is low enough, more keep going till the end.

there's an italian dish called *saltimbocca*, which means "leap into the mouth." my goal when writing might be called *saltintesta*: the ideas leap into your head and you barely notice the words that got them there.

it's too much to hope that writing could ever be pure ideas. you might not even want it to be. but for most writers, most of the time, that's the goal to aim for. the gap between most writing and pure ideas is not filled with poetry.

plus it's more considerate to write simply. when you write in a fancy way to impress people, you're making them do extra work just so you can seem cool. it's like trailing a long train behind you that readers have to carry.

and remember, if you're writing in english, that a lot of your readers won't be native english speakers. their understanding of ideas may be way ahead of their understanding of english. so you can't assume that writing about a difficult topic means you can use difficult words.

of course, fancy writing doesn't just conceal ideas. it can also conceal the lack of them. that's why some people write that way, to conceal the fact that they have nothing to say. whereas writing simply keeps you honest. if you say nothing simply, it will be obvious to everyone, including you.

simple writing also lasts better. people reading your stuff in the future will be in much the same position as people from other countries reading it today. the culture and the language will have changed. it's not vain to care about that, any more than it's vain for a woodworker to build a chair to last.

indeed, lasting is not merely an accidental quality of chairs, or writing. it's a sign you did a good job.

but although these are all real advantages of writing simply, none of them are why i do it. the main reason i write simply is that it offends me not to. when i write a sentence that seems too complicated, or that uses unnecessarily intellectual words, it doesn't seem fancy to me. it seems clumsy.

there are of course times when you want to use a complicated sentence or fancy word for effect. but you should never do it by accident.

the other reason my writing ends up being simple is the way i do it. i write the first draft fast, then spend days editing it, trying to get everything just right. much of this editing is cutting, and that makes simple writing even simpler.

here's a simple trick for getting more people to read what you write: write in spoken language.

something comes over most people when they start writing. they write in a different language than they'd use if they were talking to a friend. the sentence structure and even the words are different. no one uses "pen" as a verb in spoken english. you'd feel like an idiot using "pen" instead of "write" in a conversation with a friend.

the last straw for me was a sentence i read a couple days ago:

> the mercurial spaniard himself declared: "after altamira, all is decadence."

it's from neil oliver's *a history of ancient britain*. i feel bad making an example of this book, because it's no worse than lots of others. but just imagine calling picasso "the mercurial spaniard" when talking to a friend. even one sentence of this would raise eyebrows in conversation. and yet people write whole books of it.

ok, so written and spoken language are different. does that make written language worse?

if you want people to read and understand what you write, yes. written language is more complex, which makes it more work to read. it's also more formal and distant, which gives the reader's attention permission to drift. but perhaps worst of all, the complex sentences and fancy words give you, the writer, the false impression that you're saying more than you actually are.

you don't need complex sentences to express complex ideas. when specialists in some abstruse topic talk to one another about ideas in their field, they don't use sentences any more complex than they do when talking about what to have for lunch. they use different words, certainly. but even those they use no more than necessary. and in my experience, the harder the subject, the more informally experts speak. partly, i think, because they have less to prove, and partly because the harder the ideas you're talking about, the less you can afford to let language get in the way.

informal language is the athletic clothing of ideas.

i'm not saying spoken language always works best. poetry is as much music as text, so you can say things you wouldn't say in conversation. and there are a handful of writers who can get away with using fancy language in prose. and then of course there are cases where writers don't want to make it easy to understand what they're saying—in corporate announcements of bad news, for example, or at the more bogus end of the humanities. but for nearly everyone else, spoken language is better.

it seems to be hard for most people to write in spoken language. so perhaps the best solution is to write your first draft the way you usually would, then afterward look at each sentence and ask "is this the way i'd say this if i were talking to a friend?" if it isn't, imagine what you would say, and use that instead. after a while this filter will start to operate as you write. when you write something you wouldn't say, you'll hear the clank as it hits the page.

before i publish a new essay, i read it out loud and fix everything that doesn't sound like conversation. i even fix bits that are phonetically awkward; i don't know if that's necessary, but it doesn't cost much.

this trick may not always be enough. i've seen writing so far removed from spoken language that it couldn't be fixed sentence by sentence. for cases like that there's a more drastic solution. after writing the first draft, try explaining to a friend what you just wrote. then replace the draft with what you said to your friend.

people often tell me how much my essays sound like me talking. the fact that this seems worthy of comment shows how rarely people manage to write in spoken language. otherwise everyone's writing

i'm usually reluctant to make predictions about technology, but i feel fairly confident about this one: in a couple decades there won't be many people who can write.

one of the strangest things you learn if you're a writer is how many people have trouble writing. doctors know how many people have a mole they're worried about; people who are good at setting up computers know how many people aren't; writers know how many people need help writing.

the reason so many people have trouble writing is that it's fundamentally difficult. to write well you have to think clearly, and thinking clearly is hard.

and yet writing pervades many jobs, and the more prestigious the job, the more writing it tends to require.

these two powerful opposing forces, the pervasive expectation of writing and the irreducible difficulty of doing it, create enormous pressure. this is why eminent professors often turn out to have resorted to plagiarism. the most striking thing to me about these cases is the pettiness of the thefts. the stuff they steal is usually the most mundane boilerplate — the sort of thing that anyone who was even halfway decent at writing could turn out with no effort at all. which means they're not even halfway decent at writing.

till recently there was no convenient escape valve for the pressure created by these opposing forces. you could pay someone to write for you, like jfk, or plagiarize, like mlk, but if you couldn't buy or steal words, you had to write them yourself. and as a result nearly everyone who was expected to write had to learn how.

not anymore. ai has blown this world open. almost all pressure to write has dissipated. you can have ai do it for you, both in school and at work.

the result will be a world divided into writes and write-nots. there will still be some people who can write. some of us like it. but the middle ground between those who are good at writing and those who can't write at all will disappear. instead of good writers, ok writers, and people who can't write, there will just be good writers and people who can't write.

is that so bad? isn't it common for skills to disappear when technology makes them obsolete? there aren't many blacksmiths left, and it doesn't seem to be a problem.

yes, it's bad. the reason is something i mentioned earlier: writing is thinking. in fact there's a kind of thinking that can only be done by writing. you can't make this point better than leslie lamport did:

> if you're thinking without writing, you only think you're thinking.

so a world divided into writes and write-nots is more dangerous than it sounds. it will be a world of thinks and think-nots. i know which half i want to be in, and i bet you do too.

this situation is not unprecedented. in preindustrial times most people's jobs made them strong. now if you want to be strong, you work out. so there are still strong people, but only those who choose to be.

it will be the same with writing. there will still be smart people, but only those who choose to be.

thanks to jessica livingston, ben miller, and robert morris for reading drafts of this.

writing about something, even something you know well, usually shows you that you didn't know it as well as you thought. putting ideas into words is a severe test. the first words you choose are usually wrong; you have to rewrite sentences over and over to get them exactly right. and your ideas won't just be imprecise, but incomplete too. half the ideas that end up in an essay will be ones you thought of while you were writing it. indeed, that's why i write them.

once you publish something, the convention is that whatever you wrote was what you thought before you wrote it. these were your ideas, and now you've expressed them. but you know this isn't true. you know that putting your ideas into words changed them. and not just the ideas you published. presumably there were others that turned out to be too broken to fix, and those you discarded instead.

it's not just having to commit your ideas to specific words that makes writing so exacting. the real test is reading what you've written. you have to pretend to be a neutral reader who knows nothing of what's in your head, only what you wrote. when he reads what you wrote, does it seem correct? does it seem complete? if you make an effort, you can read your writing as if you were a complete stranger, and when you do the news is usually bad. it takes me many cycles before i can get an essay past the stranger. but the stranger is rational, so you always can, if you ask him what he needs. if he's not satisfied because you failed to mention x or didn't qualify some sentence sufficiently, then you mention x or add more qualifications. happy now? it may cost you some nice sentences, but you have to resign yourself to that. you just have to make them as good as you can and still satisfy the stranger.

this much, i assume, won't be that controversial. i think it will accord with the experience of anyone who has tried to write about anything nontrivial. there may exist people whose thoughts are so perfectly formed that they just flow straight into words. but i've never known anyone who could do this, and if i met someone who said they could, it would seem evidence of their limitations rather than their ability. indeed, this is a trope in movies: the guy who claims to have a plan for doing some difficult thing, and who when questioned further, taps his head and says "it's all up here." everyone watching the movie knows what that means. at best the plan is vague and incomplete. very likely there's some undiscovered flaw that invalidates it completely. at best it's a plan for a plan.

in precisely defined domains it's possible to form complete ideas in your head. people can play chess in their heads, for example. and mathematicians can do some amount of math in their heads, though they don't seem to feel sure of a proof over a certain length till they write it down. but this only seems possible with ideas you can express in a formal language. [1] arguably what such people are doing is putting ideas into words in their heads. i can to some extent write essays in my head. i'll sometimes think of a paragraph while walking or lying in bed that survives nearly unchanged in the final version. but really i'm writing when i do this. i'm doing the mental part of writing; my fingers just aren't moving as i do it. [2]

you can know a great deal about something without writing about it. can you ever know so much that you wouldn't learn more from trying to explain what you know? i don't think so. i've written about at least two subjects i know well — lisp hacking and startups — and in both cases i learned a lot from writing about them. in both cases there were things i didn't consciously realize till i had to explain them. and i don't think my experience was anomalous. a great deal of knowledge is unconscious, and experts have if anything a higher proportion of unconscious knowledge than beginners.

i'm not saying that writing is the best way to explore all ideas. if you have ideas about architecture, presumably the best way to explore them is to build actual buildings. what i'm saying is that however much you learn from exploring ideas in other ways, you'll still learn new things from writing about them.

what should an essay be? many people would say persuasive. that's what a lot of us were taught essays should be. but i think we can aim for something more ambitious: that an essay should be useful.

to start with, that means it should be correct. but it's not enough merely to be correct. it's easy to make a statement correct by making it vague. that's a common flaw in academic writing, for example. if you know nothing at all about an issue, you can't go wrong by saying that the issue is a complex one, that there are many factors to be considered, that it's a mistake to take too simplistic a view of it, and so on.

though no doubt correct, such statements tell the reader nothing. useful writing makes claims that are as strong as they can be made without becoming false.

for example, it's more useful to say that pike's peak is near the middle of colorado than merely somewhere in colorado. but if i say it's in the exact middle of colorado, i've now gone too far, because it's a bit east of the middle.

precision and correctness are like opposing forces. it's easy to satisfy one if you ignore the other. the converse of vaporous academic writing is the bold, but false, rhetoric of demagogues. useful writing is bold, but true.

it's also two other things: it tells people something important, and that at least some of them didn't already know.

telling people something they didn't know doesn't always mean surprising them. sometimes it means telling them something they knew unconsciously but had never put into words. in fact those may be the more valuable insights, because they tend to be more fundamental.

let's put them all together. useful writing tells people something true and important that they didn't already know, and tells them as unequivocally as possible.

notice these are all a matter of degree. for example, you can't expect an idea to be novel to everyone. any insight that you have will probably have already been had by at least one of the world's 7 billion people. but it's sufficient if an idea is novel to a lot of readers.

ditto for correctness, importance, and strength. in effect the four components are like numbers you can multiply together to get a score for usefulness. which i realize is almost awkwardly reductive, but nonetheless true.

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how can you ensure that the things you say are true and novel and important? believe it or not, there is a trick for doing this. i learned it from my friend robert morris, who has a horror of saying anything dumb. his trick is not to say anything unless he's sure it's worth hearing. this makes it hard to get opinions out of him, but when you do, they're usually right.

translated into essay writing, what this means is that if you write a bad sentence, you don't publish it. you delete it and try again. often you abandon whole branches of four or five paragraphs. sometimes a whole essay.

you can't ensure that every idea you have is good, but you can ensure that every one you publish is, by simply not publishing the ones that aren't.

in the sciences, this is called publication bias, and is considered bad. when some hypothesis you're exploring gets inconclusive results, you're supposed to tell people about that too. but with essay writing, publication bias is the way to go.

as you refine an idea, you're pushing in the direction of less qualification. but you can rarely get it down to zero. sometimes you don't even want to, if it's a side point and a fully refined version would be too long.

some say that qualifications weaken writing. for example, that you should never begin a sentence in an essay with "i think," because if you're saying it, then of course you think it. and it's true that "i think x" is a weaker statement than simply "x." which is exactly why you need "i think." you need it to express your degree of certainty.

but qualifications are not scalars. they're not just experimental error. there must be 50 things they can express: how broadly something applies, how you know it, how happy you are it's so, even how it could be falsified. i'm not going to try to explore the structure of qualification here. it's probably more complex than the whole topic of writing usefully. instead i'll just give you a practical tip: don't underestimate qualification. it's an important skill in its own right, not just a sort of tax you have to pay in order to avoid saying things that are false. so learn and use its full range. it may not be fully half of having good ideas, but it's part of having them.

there's one other quality i aim for in essays: to say things as simply as possible. but i don't think this is a component of usefulness. it's more a matter of consideration for the reader. and it's a practical aid in getting things right; a mistake is more obvious when expressed in simple language. but i'll admit that the main reason i write simply is not for the reader's sake or because it helps get things right, but because it bothers me to use more or fancier words than i need to. it seems inelegant, like a program that's too long.

i realize florid writing works for some people. but unless you're sure you're one of them, the best advice is to write as simply as you can.

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i believe the formula i've given you, importance + novelty + correctness + strength, is the recipe for a good essay. but i should warn you that it's also a recipe for making people mad.

the root of the problem is novelty. when you tell people something they didn't know, they don't always thank you for it. sometimes the reason people don't know something is because they don't want to know it. usually because it contradicts some cherished belief. and indeed, if you're looking for novel ideas, popular but mistaken beliefs are a good place to find them. every popular mistaken belief creates a dead zone of ideas around it that are relatively unexplored because they contradict it.

the strength component just makes things worse. if there's anything that annoys people more than having their cherished assumptions contradicted, it's having them flatly contradicted.

plus if you've used the morris technique, your writing will seem quite confident. perhaps offensively confident, to people who disagree with you. the reason you'll seem confident is that you are confident: you've cheated, by only publishing the things you're sure of. it will seem to people who try to disagree with you that you never admit you're wrong. in fact you constantly admit you're wrong. you just do it before publishing instead of after.

and if your writing is as simple as possible, that just makes things worse. brevity is the diction of command. if you watch someone delivering unwelcome news from a position of inferiority, you'll notice they tend to use lots of words, to soften the blow. whereas to be short with someone is more or less to be rude to them.

incidentally, steve wozniak did the same thing. in high school he designed computers on paper for fun. he couldn't build them because he couldn't afford the components. but when intel launched 4k drams in 1975, he was ready.

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how many essays are there left to write though? the answer to that question is probably the most exciting thing i've learned about essay writing. nearly all of them are left to write.

although the essay is an old form, it hasn't been assiduously cultivated. in the print era, publication was expensive, and there wasn't enough demand for essays to publish that many. you could publish essays if you were already well known for writing something else, like novels. or you could write book reviews that you took over to express your own ideas. but there was not really a direct path to becoming an essayist. which meant few essays got written, and those that did tended to be about a narrow range of subjects.

now, thanks to the internet, there's a path. anyone can publish essays online. you start in obscurity, perhaps, but at least you can start. you don't need anyone's permission.

it sometimes happens that an area of knowledge sits quietly for years, till some change makes it explode. cryptography did this to number theory. the internet is doing it to the essay.

the exciting thing is not that there's a lot left to write, but that there's a lot left to discover. there's a certain kind of idea that's best discovered by writing essays. if most essays are still unwritten, most such ideas are still undiscovered.

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notes

1 put railings on the balconies, but don't put bars on the windows.

2 even now i sometimes write essays that are not meant for publication. i wrote several to figure out what y combinator should do, and they were really helpful.

thanks to trevor blackwell, daniel gackle, jessica livingston, and robert morris for reading drafts of this.

there are two senses in which writing can be good: it can sound good, and the ideas can be right. it can have nice, flowing sentences, and it can draw correct conclusions about important things. it might seem as if these two kinds of good would be unrelated, like the speed of a car and the color it's painted. and yet i don't think they are. i think writing that sounds good is more likely to be right.

so here we have the most exciting kind of idea: one that seems both preposterous and true. let's examine it. how can this possibly be true?

i know it's true from writing. you can't simultaneously optimize two unrelated things; when you push one far enough, you always end up sacrificing the other. and yet no matter how hard i push, i never find myself having to choose between the sentence that sounds best and the one that expresses an idea best. if i did, it would be frivolous to care how sentences sound. but in practice it feels the opposite of frivolous. fixing sentences that sound bad seems to help get the ideas right. [1]

by right i mean more than just true. getting the ideas right means developing them well — drawing the conclusions that matter most, and exploring each one to the right level of detail. so getting the ideas right is not just a matter of saying true things, but saying the right true things.

how could trying to make sentences sound good help you do that? the clue to the answer is something i noticed 30 years ago when i was doing the layout for my first book. sometimes when you're laying out text you have bad luck. for example, you get a section that runs one line longer than the page. i don't know what ordinary typesetters do in this situation, but what i did was rewrite the section to make it a line shorter. you'd expect such an arbitrary constraint to make the writing worse. but i found, to my surprise, that it never did. i always ended up with something i liked better.

i don't think this was because my writing was especially careless. i think if you pointed to a random paragraph in anything written by anyone and told them to make it slightly shorter (or longer), they'd probably be able to come up with something better.

the best analogy for this phenomenon is when you shake a bin full of different objects. the shakes are arbitrary motions. or more precisely, they're not calculated to make any two specific objects fit more closely together. and yet repeated shaking inevitably makes the objects discover brilliantly clever ways of packing themselves. gravity won't let them become less tightly packed, so any change has to be a change for the better. [2]

so it is with writing. if you have to rewrite an awkward passage, you'll never do it in a way that makes it *less* true. you couldn't bear it, any more than gravity could bear things floating upward. so any change in the ideas has to be a change for the better.

it's obvious once you think about it. writing that sounds good is more likely to be right for the same reason that a well-shaken bin is more likely to be tightly packed. but there's something else going on as well. sounding good isn't just a random external force that leaves the ideas in an essay better off. it actually helps you to get them right.

the reason is that it makes the essay easier to read. it's less work to read writing that flows well. how does that help the writer? *because the writer is the first reader.* when i'm working on an essay, i spend far more time reading than writing. i'll reread some parts 50 or 100 times, replaying the thoughts in them and asking myself, like someone sanding a piece of wood, does anything catch? does anything feel wrong? and the easier the essay is to read, the easier it is to notice if something catches.

so yes, the two senses of good writing are connected in at least two ways. trying to make writing sound good makes you fix mistakes unconsciously, and also helps you fix them consciously; it shakes the bin

indeed, the two senses of good writing are more like two ends of the same thing. the connection between them is not a rigid one; the goodness of good writing is not a rod but a rope, with multiple overlapping connections running through it. but it's hard to move one end without moving the other. it's hard to be right without sounding right.

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notes

1 the closest thing to an exception is when you have to go back and insert a new point into the middle of something you've written. this often messes up the flow, sometimes in ways you can never quite repair. but i think the ultimate source of this problem is that ideas are tree-shaped and essays are linear. you inevitably run into difficulties when you try to cram the former into the latter. frankly it's surprising how much you can get away with. but even so you sometimes have to resort to an endnote.

2 obviously if you shake the bin hard enough the objects in it can become less tightly packed. and similarly, if you imposed some huge external constraint on your writing, like using alternating one and two syllable words, the ideas would start to suffer.

3 bizarrely enough, this happened in the writing of this very paragraph. an earlier version shared several phrases in common with the preceding paragraph, and the repetition bugged me each time i reread it. when i got annoyed enough to fix it, i discovered that the repetition reflected a problem in the underlying ideas, and i fixed both simultaneously.

thanks to jessica livingston and courtenay pipkin for reading drafts of this.

despite its title this isn't meant to be the best essay. my goal here is to figure out what the best essay would be like.

it would be well-written, but you can write well about any topic. what made it special would be what it was about.

obviously some topics would be better than others. it probably wouldn't be about this year's lipstick colors. but it wouldn't be vaporous talk about elevated themes either. a good essay has to be surprising. it has to tell people something they don't already know.

the best essay would be on the most important topic you could tell people something surprising about.

that may sound obvious, but it has some unexpected consequences. one is that science enters the picture like an elephant stepping into a rowboat. for example, darwin first described the idea of natural selection in an essay written in 1844. talk about an important topic you could tell people something surprising about. if that's the test of a great essay, this was surely the best one written in 1844. and indeed, the best possible essay at any given time would usually be one describing the most important scientific or technological discovery it was possible to make. [1]

another unexpected consequence: i imagined when i started writing this that the best essay would be fairly timeless — that the best essay you could write in 1844 would be much the same as the best one you could write now. but in fact the opposite seems to be true. it might be true that the best painting would be timeless in this sense. but it wouldn't be impressive to write an essay introducing natural selection now. the best essay *now* would be one describing a great discovery we didn't yet know about.

if the question of how to write the best possible essay reduces to the question of how to make great discoveries, then i started with the wrong question. perhaps what this exercise shows is that we shouldn't waste our time writing essays but instead focus on making discoveries in some specific domain. but i'm interested in essays and what can be done with them, so i want to see if there's some other question i could have asked.

there is, and on the face of it, it seems almost identical to the one i started with. instead of asking *what would the best essay be?* i should have asked *how do you write essays well?* though these seem only phrasing apart, their answers diverge. the answer to the first question, as we've seen, isn't really about essay writing. the second question forces it to be.

writing essays, at its best, is a way of discovering ideas. how do you do that well? how do you discover by writing?

an essay should ordinarily start with what i'm going to call a question, though i mean this in a very general sense: it doesn't have to be a question grammatically, just something that acts like one in the sense that it spurs some response.

how do you get this initial question? it probably won't work to choose some important-sounding topic at random and go at it. professional traders won't even trade unless they have what they call an *edge* — a convincing story about why in some class of trades they'll win more than they lose. similarly, you shouldn't attack a topic unless you have a way in — some new insight about it or way of approaching it.

you don't need to have a complete thesis; you just need some kind of gap you can explore. in fact, merely having questions about something other people take for granted can be edge enough.

think this is the optimal solution. you don't want the hunt for novelty and generality to be constrained in the moment. go with it and see what you get. [8]

since the initial question does constrain you, in the best case it sets an upper bound on the quality of essay you'll write. if you do as well as you possibly can on the chain of thoughts that follow from the initial question, the initial question itself is the only place where there's room for variation.

it would be a mistake to let this make you too conservative though, because you can't predict where a question will lead. not if you're doing things right, because doing things right means making discoveries, and by definition you can't predict those. so the way to respond to this situation is not to be cautious about which initial question you choose, but to write a lot of essays. essays are for taking risks.

almost any question can get you a good essay. indeed, it took some effort to think of a sufficiently unpromising topic in the third paragraph, because any essayist's first impulse on hearing that the best essay couldn't be about x would be to try to write it. but if most questions yield good essays, only some yield great ones.

can we predict which questions will yield great essays? considering how long i've been writing essays, it's alarming how novel that question feels.

one thing i like in an initial question is outrageousness. i love questions that seem naughty in some way — for example, by seeming counterintuitive or overambitious or heterodox. ideally all three. this essay is an example. writing about the best essay implies there is such a thing, which pseudo-intellectuals will dismiss as reductive, though it follows necessarily from the possibility of one essay being better than another. and thinking about how to do something so ambitious is close enough to doing it that it holds your attention.

i like to start an essay with a gleam in my eye. this could be just a taste of mine, but there's one aspect of it that probably isn't: to write a really good essay on some topic, you have to be interested in it. a good writer can write well about anything, but to stretch for the novel insights that are the raison d'etre of the essay, you have to care.

if caring about it is one of the criteria for a good initial question, then the optimal question varies from person to person. it also means you're more likely to write great essays if you care about a lot of different things. the more curious you are, the greater the probable overlap between the set of things you're curious about and the set of topics that yield great essays.

what other qualities would a great initial question have? it's probably good if it has implications in a lot of different areas. and i find it's a good sign if it's one that people think has already been thoroughly explored. but the truth is that i've barely thought about how to choose initial questions, because i rarely do it. i rarely *choose* what to write about; i just start thinking about something, and sometimes it turns into an essay.

am i going to stop writing essays about whatever i happen to be thinking about and instead start working my way through some systematically generated list of topics? that doesn't sound like much fun. and yet i want to write good essays, and if the initial question matters, i should care about it.

perhaps the answer is to go one step earlier: to write about whatever pops into your head, but try to ensure that what pops into your head is good. indeed, now that i think about it, this has to be the answer, because a mere list of topics wouldn't be any use if you didn't have edge with any of them. to start writing an essay, you need a topic plus some initial insight about it, and you can't generate those systematically. if only. [9]

because essays teach, and you can't teach people something they already know. natural selection is certainly a matter of permanent importance, but an essay explaining it couldn't have the same effect on us that it would have had on darwin's contemporaries, precisely because his ideas were so successful that everyone already knows about them. [10]

i imagined when i started writing this that the best possible essay would be timeless in the stricter, evergreen sense: that it would contain some deep, timeless wisdom that would appeal equally to aristotle and feynman. that doesn't seem to be true. but if the best possible essay wouldn't usually be timeless in this stricter sense, what would it take to write essays that were?

the answer to that turns out to be very strange: to be the evergreen kind of timeless, an essay has to be ineffective, in the sense that its discoveries aren't assimilated into our shared culture. otherwise there will be nothing new in it for the second generation of readers. if you want to surprise readers not just now but in the future as well, you have to write essays that won't stick — essays that, no matter how good they are, won't become part of what people in the future learn before they read them. [11]

i can imagine several ways to do that. one would be to write about things people never learn. for example, it's a long-established pattern for ambitious people to chase after various types of prizes, and only later, perhaps too late, to realize that some of them weren't worth as much as they thought. if you write about that, you can be confident of a conveyor belt of future readers to be surprised by it.

ditto if you write about the tendency of the inexperienced to overdo things — of young engineers to produce overcomplicated solutions, for example. there are some kinds of mistakes people never learn to avoid except by making them. any of those should be a timeless topic.

sometimes when we're slow to grasp things it's not just because we're obtuse or in denial but because we've been deliberately lied to. there are a lot of things adults lie to kids about, and when you reach adulthood, they don't take you aside and hand you a list of them. they don't remember which lies they told you, and most were implicit anyway. so contradicting such lies will be a source of surprises for as long as adults keep telling them.

sometimes it's systems that lie to you. for example, the educational systems in most countries train you to win by hacking the test. but that's not how you win at the most important real-world tests, and after decades of training, this is hard for new arrivals in the real world to grasp. helping them overcome such institutional lies will work as long as the institutions remain broken. [12]

another recipe for timelessness is to write about things readers already know, but in much more detail than can be transmitted culturally. "everyone knows," for example, that it can be rewarding to have kids. but till you have them you don't know precisely what forms that takes, and even then much of what you know you may never have put into words.

i've written about all these kinds of topics. but i didn't do it in a deliberate attempt to write essays that were timeless in the stricter sense. and indeed, the fact that this depends on one's ideas not sticking suggests that it's not worth making a deliberate attempt to. you should write about topics of timeless importance, yes, but if you do such a good job that your conclusions stick and future generations find your essay obvious instead of novel, so much the better. you've crossed into darwin territory.

writing about topics of timeless importance is an instance of something even more general, though: breadth of applicability. and there are more kinds of breadth than chronological — applying to lots of different fields, for example. so breadth is the ultimate aim.

it and then just reproduce it again.

strictly speaking it's the graph of ideas that would be highly connected, not the space, but that usage would confuse people who don't know graph theory, whereas people who do know it will get what i mean if i say "space".

8 too far doesn't depend just on the distance from the original topic. it's more like that distance divided by the value of whatever i've discovered in the subtree.

9 or can you? i should try writing about this. even if the chance of succeeding is small, the expected value is huge.

10 there was a vogue in the 20th century for saying that the purpose of art was also to teach. some artists tried to justify their work by explaining that their goal was not to produce something good, but to challenge our preconceptions about art. and to be fair, art can teach somewhat. the ancient greeks' naturalistic sculptures represented a new idea, and must have been extra exciting to contemporaries on that account. but they still look good to us.

11 bertrand russell caused huge controversy in the early 20th century with his ideas about "trial marriage." but they make boring reading now, because they prevailed. "trial marriage" is what we call "dating."

12 if you'd asked me 10 years ago, i'd have predicted that schools would continue to teach hacking the test for centuries. but now it seems plausible that students will soon be taught individually by ais, and that exams will be replaced by ongoing, invisible micro-assessments.

thanks to sam altman, trevor blackwell, jessica livingston, robert morris, courtenay pipkin, and harj taggar for reading drafts of this.

an essay has to tell people something they don't already know. but there are three different reasons people might not know something, and they yield three very different kinds of essays.

one reason people won't know something is if it's not important to know. that doesn't mean it will make a bad essay. for example, you might write a good essay about a particular model of car. readers would learn something from it. it would add to their picture of the world. for a handful of readers it might even spur some kind of epiphany. but unless this is a very unusual car it's not critical for everyone to know about it. [1]

if something isn't important to know, there's no answer to the question of why people don't know it. not knowing random facts is the default. but if you're going to write about things that *are* important to know, you have to ask why your readers don't already know them. is it because they're inexperienced, or because they're obtuse?

so the three reasons readers might not already know what you tell them are (a) that it's not important, (b) that they're obtuse, or (c) that they're inexperienced.

the reason i did this breakdown was to get at the following fact, which might have seemed controversial if i'd led with it, but should be obvious now. if you're writing for smart people about important things, you're writing for the young.

or more precisely, that's where you'll have the most effect. whatever you say should also be at least somewhat novel to you, however old you are. it's not an essay otherwise, because an essay is something you write to figure something out. but whatever you figure out will presumably be more of a surprise to younger readers than it is to you.

there's a continuum of surprise. at one extreme, something you read can change your whole way of thinking. *the selfish gene* did this to me. it was like suddenly seeing the other interpretation of an ambiguous image: you can treat genes rather than organisms as the protagonists, and evolution becomes easier to understand when you do. at the other extreme, writing merely puts into words something readers were already thinking — or thought they were.

the impact of an essay is how much it changes readers' thinking multiplied by the importance of the topic. but it's hard to do well at both. it's hard to have big new ideas about important topics. so in practice there's a tradeoff: you can change readers' thinking a lot about moderately important things, or change it a little about very important ones. but with younger readers the tradeoff shifts. there's more room to change their thinking, so there's a bigger payoff for writing about important things.

the tradeoff isn't a conscious one, at least not for me. it's more like a kind of gravitational field that writers work in. but every essayist works in it, whether they realize it or not.

this seems obvious once you state it, but it took me a long time to understand. i knew i wanted to write for smart people about important topics. i noticed empirically that i seemed to be writing for the young. but it took me years to understand that the latter was an automatic consequence of the former. in fact i only really figured it out as i was writing this essay.

now that i know it, should i change anything? i don't think so. in fact seeing the shape of the field that writers work in has reminded me that i'm not optimizing for returns in it. i'm not trying to surprise readers of any particular age; i'm trying to surprise myself.

the way i usually decide what to write about is by following curiosity. i notice something new and dig into it. it would probably be a mistake to change that. but seeing the shape of the essay field has set me

remember the essays you had to write in high school? topic sentence, introductory paragraph, supporting paragraphs, conclusion. the conclusion being, say, that ahab in *moby dick* was a christ-like figure.

oy. so i'm going to try to give the other side of the story: what an essay really is, and how you write one. or at least, how i write one.

mods

the most obvious difference between real essays and the things one has to write in school is that real essays are not exclusively about english literature. certainly schools should teach students how to write. but due to a series of historical accidents the teaching of writing has gotten mixed together with the study of literature. and so all over the country students are writing not about how a baseball team with a small budget might compete with the yankees, or the role of color in fashion, or what constitutes a good dessert, but about symbolism in dickens.

with the result that writing is made to seem boring and pointless. who cares about symbolism in dickens? dickens himself would be more interested in an essay about color or baseball.

how did things get this way? to answer that we have to go back almost a thousand years. around 1100, europe at last began to catch its breath after centuries of chaos, and once they had the luxury of curiosity they rediscovered what we call "the classics." the effect was rather as if we were visited by beings from another solar system. these earlier civilizations were so much more sophisticated that for the next several centuries the main work of european scholars, in almost every field, was to assimilate what they knew.

during this period the study of ancient texts acquired great prestige. it seemed the essence of what scholars did. as european scholarship gained momentum it became less and less important; by 1350 someone who wanted to learn about science could find better teachers than aristotle in his own era.[1] but schools change slower than scholarship. in the 19th century the study of ancient texts was still the backbone of the curriculum.

the time was then ripe for the question: if the study of ancient texts is a valid field for scholarship, why not modern texts? the answer, of course, is that the original raison d'etre of classical scholarship was a kind of intellectual archaeology that does not need to be done in the case of contemporary authors. but for obvious reasons no one wanted to give that answer. the archaeological work being mostly done, it implied that those studying the classics were, if not wasting their time, at least working on problems of minor importance.

and so began the study of modern literature. there was a good deal of resistance at first. the first courses in english literature seem to have been offered by the newer colleges, particularly american ones. dartmouth, the university of vermont, amherst, and university college, london taught english literature in the 1820s.

but harvard didn't have a professor of english literature until 1876, and oxford not till 1885. (oxford had a chair of chinese before it had one of english.)[2]

what tipped the scales, at least in the us, seems to have been the idea that professors should do research as well as teach. this idea (along with the phd, the department, and indeed the whole concept of the modern university) was imported from germany in the late 19th century. beginning at johns hopkins in 1876, the new model spread rapidly.

writing was one of the casualties. colleges had long taught english composition. but how do you do research on composition? the professors who taught math could be required to do original math, the

usually be fixed by cutting. but i don't try to fix the unconvincing bits by arguing more cleverly. i need to talk the matter over.

at the very least i must have explained something badly. in that case, in the course of the conversation i'll be forced to come up a with a clearer explanation, which i can just incorporate in the essay. more often than not i have to change what i was saying as well. but the aim is never to be convincing per se. as the reader gets smarter, convincing and true become identical, so if i can convince smart readers i must be near the truth.

the sort of writing that attempts to persuade may be a valid (or at least inevitable) form, but it's historically inaccurate to call it an essay. an essay is something else.

trying

to understand what a real essay is, we have to reach back into history again, though this time not so far. to michel de montaigne, who in 1580 published a book of what he called "essais." he was doing something quite different from what lawyers do, and the difference is embodied in the name. *essayer* is the french verb meaning "to try" and an *essai* is an attempt. an essay is something you write to try to figure something out.

figure out what? you don't know yet. and so you can't begin with a thesis, because you don't have one, and may never have one. an essay doesn't begin with a statement, but with a question. in a real essay, you don't take a position and defend it. you notice a door that's ajar, and you open it and walk in to see what's inside.

if all you want to do is figure things out, why do you need to write anything, though? why not just sit and think? well, there precisely is montaigne's great discovery. expressing ideas helps to form them. indeed, helps is far too weak a word. most of what ends up in my essays i only thought of when i sat down to write them. that's why i write them.

in the things you write in school you are, in theory, merely explaining yourself to the reader. in a real essay you're writing for yourself. you're thinking out loud.

but not quite. just as inviting people over forces you to clean up your apartment, writing something that other people will read forces you to think well. so it does matter to have an audience. the things i've written just for myself are no good. they tend to peter out. when i run into difficulties, i find i conclude with a few vague questions and then drift off to get a cup of tea.

many published essays peter out in the same way. particularly the sort written by the staff writers of newsmagazines. outside writers tend to supply editorials of the defend-a-position variety, which make a beeline toward a rousing (and foreordained) conclusion. but the staff writers feel obliged to write something "balanced." since they're writing for a popular magazine, they start with the most radioactively controversial questions, from which-- because they're writing for a popular magazine-- they then proceed to recoil in terror. abortion, for or against? this group says one thing. that group says another. one thing is certain: the question is a complex one. (but don't get mad at us. we didn't draw any conclusions.)

the river

questions aren't enough. an essay has to come up with answers. they don't always, of course. sometimes you start with a promising question and get nowhere. but those you don't publish. those are like experiments that get inconclusive results. an essay you publish ought to tell the reader something he didn't already know.

so if you want to write essays, you need two ingredients: a few topics you've thought about a lot, and some ability to ferret out the unexpected.

what should you think about? my guess is that it doesn't matter-- that anything can be interesting if you get deeply enough into it. one possible exception might be things that have deliberately had all the variation sucked out of them, like working in fast food. in retrospect, was there anything interesting about working at baskin-robbins? well, it was interesting how important color was to the customers. kids a certain age would point into the case and say that they wanted yellow. did they want french vanilla or lemon? they would just look at you blankly. they wanted yellow. and then there was the mystery of why the perennial favorite pralines 'n' cream was so appealing. (i think now it was the salt.)

and the difference in the way fathers and mothers bought ice cream for their kids: the fathers like benevolent kings bestowing largesse, the mothers harried, giving in to pressure. so, yes, there does seem to be some material even in fast food.

i didn't notice those things at the time, though. at sixteen i was about as observant as a lump of rock. i can see more now in the fragments of memory i preserve of that age than i could see at the time from having it all happening live, right in front of me.

observation

so the ability to ferret out the unexpected must not merely be an inborn one. it must be something you can learn. how do you learn it?

to some extent it's like learning history. when you first read history, it's just a whirl of names and dates. nothing seems to stick. but the more you learn, the more hooks you have for new facts to stick onto-- which means you accumulate knowledge at an exponential rate. once you remember that normans conquered england in 1066, it will catch your attention when you hear that other normans conquered southern italy at about the same time. which will make you wonder about normandy, and take note when a third book mentions that normans were not, like most of what is now called france, tribes that flowed in as the roman empire collapsed, but vikings (norman = north man) who arrived four centuries later in 911. which makes it easier to remember that dublin was also established by vikings in the 840s. etc, etc squared.

collecting surprises is a similar process. the more anomalies you've seen, the more easily you'll notice new ones. which means, oddly enough, that as you grow older, life should become more and more surprising. when i was a kid, i used to think adults had it all figured out. i had it backwards. kids are the ones who have it all figured out. they're just mistaken.

when it comes to surprises, the rich get richer. but (as with wealth) there may be habits of mind that will help the process along. it's good to have a habit of asking questions, especially questions beginning with why. but not in the random way that three year olds ask why. there are an infinite number of questions. how do you find the fruitful ones?

i find it especially useful to ask why about things that seem wrong. for example, why should there be a connection between humor and misfortune? why do we find it funny when a character, even one we like, slips on a banana peel? there's a whole essay's worth of surprises there for sure.

if you want to notice things that seem wrong, you'll find a degree of skepticism helpful. i take it as an axiom that we're only achieving 1% of what we could. this helps counteract the rule that gets beaten into our heads as children: that things are the way they are because that is how things have to be. for example, everyone i've talked to while writing this essay felt the same about english classes-- that the

see what you can extract from a frivolous question? if there's one piece of advice i would give about writing essays, it would be: don't do as you're told. don't believe what you're supposed to. don't write the essay readers expect; one learns nothing from what one expects. and don't write the way they taught you to in school.

the most important sort of disobedience is to write essays at all. fortunately, this sort of disobedience shows signs of becoming rampant. it used to be that only a tiny number of officially approved writers were allowed to write essays. magazines published few of them, and judged them less by what they said than who wrote them; a magazine might publish a story by an unknown writer if it was good enough, but if they published an essay on x it had to be by someone who was at least forty and whose job title had x in it. which is a problem, because there are a lot of things insiders can't say precisely because they're insiders.

the internet is changing that. anyone can publish an essay on the web, and it gets judged, as any writing should, by what it says, not who wrote it. who are you to write about x? you are whatever you wrote.

popular magazines made the period between the spread of literacy and the arrival of tv the golden age of the short story. the web may well make this the golden age of the essay. and that's certainly not something i realized when i started writing this.

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notes

1 i'm thinking of oresme (c. 1323-82). but it's hard to pick a date, because there was a sudden drop-off in scholarship just as europeans finished assimilating classical science. the cause may have been the plague of 1347; the trend in scientific progress matches the population curve.

2 parker, william r. "where do college english departments come from?" *college english* 28 (1966-67), pp. 339-351. reprinted in gray, donald j. (ed). *the department of english at indiana university bloomington 1868-1970.* indiana university publications.

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3 i'm compressing the story a bit. at first literature took a back seat to philology, which (a) seemed more serious and (b) was popular in germany, where many of the leading scholars of that generation had been trained.

in some cases the writing teachers were transformed *in situ* into english professors. francis james child, who had been boylston professor of rhetoric at harvard since 1851, became in 1876 the university's first professor of english.

4 parker, *op. cit.*, p. 25.

5 the undergraduate curriculum or *trivium* (whence "trivial") consisted of latin grammar, rhetoric, and logic. candidates for masters' degrees went on to study the *quadrivium* of arithmetic, geometry, music, and astronomy. together these were the seven liberal arts.

the study of rhetoric was inherited directly from rome, where it was considered the most important subject. it would not be far from the truth to say that education in the classical world meant training landowners' sons to speak well enough to defend their interests in political and legal disputes.

since i was about 9 i've been puzzled by the apparent contradiction between being made of matter that behaves in a predictable way, and the feeling that i could choose to do whatever i wanted. at the time i had a self-interested motive for exploring the question. at that age (like most succeeding ages) i was always in trouble with the authorities, and it seemed to me that there might possibly be some way to get out of trouble by arguing that i wasn't responsible for my actions. i gradually lost hope of that, but the puzzle remained: how do you reconcile being a machine made of matter with the feeling that you're free to choose what you do? [1]

the best way to explain the answer may be to start with a slightly wrong version, and then fix it. the wrong version is: you can do what you want, but you can't want what you want. yes, you can control what you do, but you'll do what you want, and you can't control that.

the reason this is mistaken is that people do sometimes change what they want. people who don't want to want something — drug addicts, for example — can sometimes make themselves stop wanting it. and people who want to want something — who want to like classical music, or broccoli — sometimes succeed.

so we modify our initial statement: you can do what you want, but you can't want to want what you want.

that's still not quite true. it's possible to change what you want to want. i can imagine someone saying "i decided to stop wanting to like classical music." but we're getting closer to the truth. it's rare for people to change what they want to want, and the more "want to"s we add, the rarer it gets.

we can get arbitrarily close to a true statement by adding more "want to"s in much the same way we can get arbitrarily close to 1 by adding more 9s to a string of 9s following a decimal point. in practice three or four "want to"s must surely be enough. it's hard even to envision what it would mean to change what you want to want to want to want, let alone actually do it.

so one way to express the correct answer is to use a regular expression. you can do what you want, but there's some statement of the form "you can't (want to)* want what you want" that's true. ultimately you get back to a want that you don't control. [2]

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notes

1 i didn't know when i was 9 that matter might behave randomly, but i don't think it affects the problem much. randomness destroys the ghost in the machine as effectively as determinism.

2 if you don't like using an expression, you can make the same point using higher-order desires: there is some n such that you don't control your nth-order desires.

thanks to trevor blackwell, jessica livingston, robert morris, and michael nielsen for reading drafts of this.

(someone fed my essays into gpt to make something that could answer questions based on them, then asked it where good ideas come from. the answer was ok, but not what i would have said. this is what i would have said.)

the way to get new ideas is to notice anomalies: what seems strange, or missing, or broken? you can see anomalies in everyday life (much of standup comedy is based on this), but the best place to look for them is at the frontiers of knowledge.

knowledge grows fractally. from a distance its edges look smooth, but when you learn enough to get close to one, you'll notice it's full of gaps. these gaps will seem obvious; it will seem inexplicable that no one has tried x or wondered about y. in the best case, exploring such gaps yields whole new fractal buds.